CHITKARA UNIVERSITY

Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

Proactive Prediction of UPI Transaction Failure Using Machine Learning on Synthetic Data

Priyam Aggarwal

2210992098

Supervised By

Dr. Ajay Kumar

Associate Professor

Department of Computer Science and Engineering,

Chitkara University, Punjab

Introduction

- UPI is India's leading real-time payment system, handling billions of transactions monthly.
- Despite its scale, transaction failures still occur due to bank server overload, network instability, and peak-hour congestion.
- Existing systems only detect failures after they happen no proactive prediction exists.
- This research proposes a machine learning approach to predict UPI transaction failures before execution.
- A synthetic Kaggle dataset is used to simulate real-world scenarios while maintaining data privacy.
- Models used: Random Forest and XGBoost both well-suited for structured financial data classification.

Problem Statement

- UPI transaction failures degrade user experience and lead to repeated attempts, increasing system load.
- Failures occur due to bank server downtime, poor network conditions, and high concurrent transaction volumes.
- No proactive mechanism exists to warn users before a likely-to-fail transaction is submitted.
- Real banking data is sensitive and unavailable making it difficult to train predictive models on actual data.
- Most existing ML research in payments focuses on fraud detection, not reliability or failure prediction.
- There is a gap: failure patterns in transaction metadata have not been systematically modeled for prediction.

Proposed solution

- A machine learning-based predictive layer integrated alongside existing UPI systems.
- Uses a synthetic dataset (Kaggle) to train classifiers ensuring privacy and ethical compliance.
- Extracts features: transaction hour, day of week, sender bank, receiver bank, and amount.
- Two models trained and compared: Random Forest and XGBoost.
- Prediction output: SUCCESS / FAILURE likelihood before the user submits the transaction.
- Acts as a decision support tool does not replace the actual transaction engine.

Random Forest

Ensemble, robust to overfitting

XGBoost

Gradient boosting, high accuracy

Synthetic Data

Privacy-preserving, Kaggle

Methodology

CHITKARA UNIVERSITY

1
Data Collection
Synthetic CSV (Kaggle)

2
Preprocessing
Timestamps, null handling, encoding

3
Feature Engineering
Hour, Day, Banks, Amount

4
Model Training
RF & XGBoost 80/20 split

5
Evaluation & Predict
Accuracy, F1, ROC-AUC

Evaluation Metrics Used:

Metric	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Purpose	Overall correctness	Correct +ve predictions	Detects actual failures	Precision–Recall balance	Classification quality

System Architecture

CHITKARA UNIVERSITY

Data Ingestion

CSV — Synthetic Kaggle Dataset

Processing Layer

Notebooks · Feature Engineering · EDA

Model Training

Random Forest · XGBoost

Evaluation & Output

Metrics · Prediction · Plots

Key Design Principles

- Predictive layer works alongside existing UPI systems, not as a replacement.
- Pattern learning from historical synthetic transactions.
- Pre-execution failure probability scoring.
- Decision support output: SUCCESS / FAILURE likelihood.
- Modular architecture each layer is independently testable.

Fututre Scope

- Integrate real-time bank server & network data for higher accuracy.
- Deploy in live UPI apps for pre-transaction failure warnings.
- Explore deep learning (LSTM, Transformers) for temporal patterns.
- Add Explainable AI (XAI) to clarify why transactions may fail.
- Build a smart payment assistant suggesting retry or bank switch.

Conclusion

- ML can proactively predict UPI failures using transaction metadata.
- Transaction hour, day, amount, and bank IDs are strong predictors.
- Both Random Forest and XGBoost achieved satisfactory performance.
- Synthetic data enables privacy-compliant model development.
- System acts as a decision-support layer improving user experience.

Thank You